SAFE BD

SAFETY • AWARENESS • FAST RESPONSE • EMERGENCY ASSISTANCE

OFFICIAL SYSTEM SPECIFICATION REPORT
Document Ref: SAFEBD-SPEC-2026-V4
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Live Platform: safebd.onrender.com

Executive Summary: SAFE bd is Bangladesh's first unified citizen safety, digital emergency dispatch, and commuter protection ecosystem. Designed to eliminate delays between citizen distress and law enforcement response, SAFE bd integrates **National 999 CAD dispatch**, **satellite GPS route monitoring**, **tamper-proof cloud evidence recording**, and **automated WhatsApp family beacons** into a high-performance, offline-capable Progressive Web Application (PWA).

1. Core System Architecture & Infrastructure Stack

SAFE bd is engineered with a fault-tolerant multi-tier architecture to guarantee uninterrupted availability even during cellular network disruption or cloud server restarts:

Layer	Technologies Deployed	Operational Capabilities & Resilience
Frontend Client	Mobile-first PWA, HTML5, CSS3, ES6+, Service Workers, Leaflet GIS	Zero-install universal compatibility across Android, iOS, and desktop browsers. Offline caching, responsive touch UI, zero-lag rendering.
Backend API & Dispatch	Node.js, Express.js, WebSockets (ws), RESTful micro-endpoints	Sub-second WebSocket telemetry, bi-directional live GPS coordinate broadcasting, rate-limiting, and cryptographic session protection.
Database & Cloud Persistence	Atomic Disk JSON, Supabase PostgreSQL, 5TB Google Drive Vault	Triple-layer failover. Local atomic disk saves, cloud PostgreSQL database replication, and massive 5TB media vault for video/audio evidence.
Production Hosting	Render Cloud Container + Local Hybrid Failover	Automatic TLS encryption, continuous GitHub CI/CD synchronization, container monitoring, and health checking endpoints.

2. Complete Feature Matrix: Emergency Distress & Rapid SOS Engine

The core SOS subsystem provides immediate, high-priority lifelines when a citizen is threatened, cornered, or experiencing medical trauma:

🚨 Instant One-Tap & Triple-Tap SOS Trigger

CRITICAL LIFELINE

High-visibility central emergency button engineered with instant haptic tactile feedback and intelligent accidental-touch prevention.

- **Triple-Tap Quick Access:** Rapidly triple-clicking the SOS circle activates full emergency mode without looking at the screen.
- **Audio Siren vs. Silent SOS:** Citizens can toggle between a high-decibel deterrent siren to scare off assailants, or a completely silent covert SOS that alerts authorities and family without escalating danger.
- **Satellite GPS Coordinates:** Real-time latitude, longitude, accuracy radius, and street-level reverse geocoding captured immediately upon tap.

📞 Automated 999 National Emergency CAD Dispatch Integration

OFFICIAL PROTOCOL

Direct bridge to Bangladesh's National Emergency Hotline 999.

- **Instant 1-Click Dial:** Initiates direct tel-protocol connection to 999 dispatchers with caller identification and auto-prepared case notes.
- **WebSocket CAD Dispatcher:** Concurrently broadcasts emergency payload to law enforcement dispatch consoles with live location updates.

📹 Covert Video & Audio Evidence Recording Vault

ANTI-TAMPER

Autonomous background evidence capture during emergencies.

- **Background Media Capture:** Records live ambient audio and camera clips covertly, preventing attackers from noticing or deleting recordings.
- **Direct 5TB Vault Upload:** Recordings bypass device storage and upload directly to SAFE bd's secure Google Drive Vault and Supabase cloud. Even if the citizen’s phone is smashed or stolen, the evidence remains intact for police prosecution.

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OFFICIAL SYSTEM SPECIFICATION REPORT
Section 3 & 4: Journey Companion & Public Directory
Platform URL: safebd.onrender.com

3. "Walk With Me" Personal Companion Guard (Commuter Safety)

Engineered for women, students, and late-night commuters traveling through high-risk urban or rural corridors:

🚗 Vehicle Registration & Dhaka Metro Plate Tracking

COMMUTER SHIELD

Prior to boarding public transit, rideshares, or rickshaws, commuters enter the vehicle's identity.

- **Bangladesh Plate Standard:** Pre-configured for official BRTA classifications (e.g. ঢাকা মেট্রো, চট্টগ্রাম মেট্রো, গ, ঘ, থ, ছ, etc.) + 6-digit registration numbers.
- **Walking Mode:** One-tap toggle for on-foot travel where no vehicle plate is required.
- **Floating Apple-Style Journey Dock:** Real-time journey bar displays remaining time, travel mode, and registered plate number throughout transit.

⏱️ Journey Safety Timer & Proximity Geofencing (10-15m Auto-Arrival)

INTELLIGENT AUTOMATION

Autonomous journey monitoring using satellite GPS coordinates.

- **Custom Safety Durations:** Commuters set predicted travel times (15 min, 30 min, 1 hour, or custom).
- **Auto-Stop Geofence:** Automatically detects when the user arrives within 10–15 meters of their designated destination, automatically disarming the safety timer and sending an arrival confirmation.
- **Automated Overdue Escalation:** If the timer expires without user confirmation, the system triggers the *"Did you reach home safely?"* safety modal and prepares automatic distress escalation.

💬 Direct WhatsApp Emergency Distress Dispatch

INSTANT FAMILY ALERT

Replaces slow SMS with instant, rich-text WhatsApp emergency dispatches.

- **1-Tap WhatsApp SOS:** Clicking emergency opens WhatsApp directly to selected guardians with formatted markdown (bold alert, timestamp, address).
- **Auto-Attached Live GPS Link:** Includes a direct Google Maps URL (`https://maps.google.com/?q=lat,lon`) showing exact coordinates.
- **Auto-Attached Vehicle Plate:** Automatically includes the car/CNG number so family and police know exactly which vehicle to intercept.
- **Universal Share-All:** Broadcasts the distress beacon to multiple WhatsApp contacts or family groups in one click.

4. Nationwide Public Safety & Emergency Directories

SAFE bd hosts offline-accessible, verified directories connecting citizens to immediate localized assistance across all 64 districts of Bangladesh:

Directory Module	Data Coverage & Reach	Citizen Action & Utility
Bangladesh Police Master Directory	**641 Police Thanas** spanning all 64 districts & 8 administrative divisions. Verified Officer-in-Charge (OC) and Duty Officer telephone lines.	Instant localized search by division and district. One-tap direct call to local thana duty desks without looking up phonebooks.
National Emergency Hospital Directory	Major public, specialized, and private medical colleges and trauma centers across all 8 divisions.	Immediate access to emergency room (ER) hotlines, ambulance dispatch desks, and 24/7 hospital admission desks.
National Specialized Hotlines	999 (National Emergency), 109 (Violence Against Women & Children), Fire Service & Civil Defense, Disaster Helpline.	High-priority instant calling buttons with zero charge and zero delay.
Blood Donor Matching Network	Citizen blood registry supporting all 8 blood groups (A+, A-, B+, B-, O+, O-, AB+, AB-).	Instant emergency blood broadcast and direct donor search for urgent surgeries and accident victims.

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Section 5, 6 & 7: Community, AI & Security
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5. Community Awareness Feed & Civic Incident Reporting

An interactive, verified civic reporting platform designed to crowdsource community safety vigilance while eliminating fake news and identity spoofing:

🛡️ Cryptographic Identity Protection & Verified Profile Badges

SECURITY SHIELD

Strict tokenized verification prevents bad actors from spoofing official authority figures or spreading panic.

- **Official Founder & CEO Badge:** Exclusively awarded to verified sessions matching cryptographic credentials (`SAAD RIZWAN`, `01316750712`). Any unauthenticated user attempting to post under the Founder's name is automatically downgraded to "Citizen Reporter".
- **Verified Citizen & Admin Badges:** Distinguishes authenticated citizen reports from general anonymous submissions.

📱 Facebook-Style Social Safety Feed & Media Vaulting

SOCIAL ENGAGEMENT

Familiar, intuitive community engagement keeping neighborhoods informed of local hazards and crime alerts.

- **Category Filtering:** Filter reports by *General Safety*, *Emergency Alert*, *Crime Alert*, or *Community Watch*.
- **High-Capacity Media Uploads:** Incident photos and videos are stored in the private 5TB Google Drive Vault to ensure low server overhead and permanent evidence retention.
- **Social Threading:** Supports real-time like counts and interactive threaded citizen comments.

6. ShurokhaAI & Multi-Functional Emergency Intelligence

🤖 ShurokhaAI 24/7 Universal Safety Companion

AI ASSISTANT

Floating, non-intrusive bilingual artificial intelligence companion.

- **Emergency First Aid & Trauma Support:** Provides instantaneous step-by-step guidance for CPR, burn management, severe bleeding control, and fracture immobilization.
- **Legal & Citizen Rights Guidance:** Explains how to file a police GD (General Diary), legal protections under Bangladesh law against harassment, and survivor rights.
- **Bilingual Conversational Interface:** Full bi-directional support for Bangla (বাংলা) and English.

7. Comprehensive Capability Comparison: Traditional Response vs. SAFE BD

Capability Domain	Traditional Citizen Experience	SAFE BD Integrated Ecosystem
Distress Alert Dispatch	Manual phone dial during high panic; victim must verbalize exact location under threat.	**1-Tap Silent or Audible SOS:** Automatically transmits satellite GPS coordinates, vehicle plate, and time in 1 second.
Commuter & Travel Safety	No automated tracking; families only notice after a traveler goes missing for hours.	**Walk With Me Companion:** Real-time journey timer, vehicle plate tracking, auto-geofence arrival, and instant WhatsApp family dispatches.
Crime Evidence Preservation	Phone destroyed or stolen by attackers; video/audio lost forever.	**5TB Cloud Vault:** Background media instantly encrypted and streamed to offsite cloud vault, un-deletable by attackers.
Police Directory Access	Citizens struggle to find local Thana OC phone numbers during late-night emergencies.	**Offline 641 Thana Directory:** Nationwide access to all 64 districts with one-touch direct dial to duty desks.

> **Verification & National Readiness:** All listed features in this specification are fully developed, tested across automated security suites, and currently deployed live at **https://safebd.onrender.com**. SAFE bd stands ready for immediate public-private partnership and national pilot integration with the Ministry of Home Affairs, ICT Division, and Bangladesh Police.